

Cloud Distributing Computing Simulation

(Node Crash Handling)

Algorithm:

Input:

Number of nodes N (e.g., 4)

A list of tasks with IDs and types

Output:

All tasks executed by alive nodes

Node status (Alive/Dead, tasks completed)

Step 1 – Initialize Nodes

Create N compute nodes

Set each node's alive flag = true

Initialize taskQueue for each node

Step 2 – Create Tasks

Define tasks: DOCUMENT, PPT, DATA_ANALYSIS, CALCULATION

Assign unique task IDs

Step 3 – Distribute Tasks Using Load Balancer

For each task:

- a. Check all alive nodes
- b. Find node with minimum tasks in queue

c. Assign the task to that node

Step 4 – Simulate Node Crash (Optional)

Randomly select a node

Set node alive = false

Print Node X crashed

Steal tasks from the crashed node

Reassign stolen tasks to other alive nodes using Step 3

Step 5 – Execute Tasks on Nodes

For each node:

- a. If node is alive:
 - i. Take tasks from taskQueue
 - ii. Execute each task (execute() method)
 - iii. Increment processedCount

Step 6 – Print Final Status

For each node, print:

Node ID

Alive / Dead

Tasks completed

Step 7 – End

Simulation complete

Output

Compile Result

DISTRIBUTED SYSTEM WITH FAULT TOLERANCE

✗ Node 3 CRASHED!

Node 1 executing Task 1 [Document Creation]

Node 1 executing Task 5 [Document Creation]

Node 1 executing Task 3 [Data Analysis]

Node 2 executing Task 2 [PPT Creation]

Node 2 executing Task 6 [PPT Creation]

Node 2 executing Task 7 [Data Analysis]

Node 4 executing Task 4 [Heavy Calculation]

Node 4 executing Task 8 [Heavy Calculation]

===== FINAL STATUS =====

Node 1 | Alive: YES | Tasks Done: 3

Node 2 | Alive: YES | Tasks Done: 3

Node 3 | Alive: NO | Tasks Done: 0

Node 4 | Alive: YES | Tasks Done: 2

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Simulation Completed Successfully

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